

SULA, Norway

WMO: 012280

Lat: 63.847N

Lon: 8.467E

Elev: 20

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	22.4	26.0	13.6	11.0	24.9	16.3	12.5	27.0	46.8	42.2	43.0	42.3	17.6	110	0.785

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.3	67.5	59.8	64.3	58.2	61.9	57.1	61.0	65.3	59.3	62.9	57.8	61.0	9.3	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.2	75.1	62.5	57.5	70.6	60.8	56.1	67.1	59.5	27.1	65.4	25.9	63.1	25.0	61.1	68.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
41.7	36.5	32.2	DB	21.1	73.4	4.4	3.2	17.9	75.7	15.4	77.6	12.9	79.4	9.6	81.7
			WB	19.1	63.8	4.1	2.6	16.1	65.7	13.7	67.2	11.4	68.7	8.5	70.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	36.8	36.2	37.7	41.8	46.5	50.9	55.5	57.1	53.8	47.3	41.6	38.6	
	DBStd	8.54	4.76	5.25	4.58	4.32	4.37	3.70	3.92	3.59	3.80	4.44	5.19	5.51	
	HDD50	2307	409	386	381	251	128	29	1	0	10	104	254	354	
	HDD65	7170	874	806	846	697	574	422	297	247	337	549	702	819	
	CDD50	615	0	0	0	4	19	57	171	220	123	20	1	0	
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0	
	CDH74	6	0	0	0	0	0	0	5	1	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	14.4	17.2	16.9	16.5	13.1	11.6	11.5	10.1	10.8	14.4	15.8	16.9	18.1	
Precipitation	PrecAvg	57.4	5.4	5.3	5.4	3.1	3.0	3.8	3.6	4.8	5.9	5.6	5.3	6.1	
	PrecMax	68.7	13.7	12.4	13.2	7.2	5.9	7.7	6.9	10.1	14.8	10.3	12.0	12.9	
	PrecMin	41.5	0.2	0.2	1.9	0.6	1.6	1.3	1.0	1.5	2.0	1.2	0.2	1.7	
	PrecStd	7.4	3.3	2.7	3.0	1.8	1.1	1.5	1.5	2.1	3.1	1.9	3.5	2.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.3	46.6	48.4	59.3	64.5	67.5	72.8	72.0	65.7	58.8	53.0	49.6	
		MCWB	44.4	43.9	42.8	50.2	53.9	57.7	62.2	61.7	58.6	52.5	49.8	46.9	
	2%	DB	45.8	45.2	46.1	52.7	58.6	62.1	67.0	68.0	62.6	56.2	51.3	48.4	
		MCWB	43.4	42.9	42.8	46.3	51.6	55.8	60.0	60.9	56.7	51.7	48.3	46.0	
	5%	DB	44.6	44.2	45.0	49.3	55.2	58.3	63.5	64.5	60.2	54.5	49.6	47.1	
		MCWB	42.5	41.8	42.2	44.8	49.7	53.7	58.4	58.9	55.5	50.7	46.6	44.6	
	10%	DB	43.4	42.8	43.8	46.9	52.2	55.8	60.9	62.1	58.5	52.9	48.2	45.6	
		MCWB	41.2	40.2	41.2	43.5	48.0	52.1	57.1	57.9	54.8	49.6	45.4	42.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.6	44.9	45.0	50.6	55.4	59.8	63.2	64.2	60.1	54.6	50.9	48.1
			MCDB	46.4	45.7	46.0	58.1	61.7	65.1	69.7	68.6	63.3	56.6	52.3	49.0
2%		WB	44.2	43.6	44.2	47.5	52.4	56.3	60.7	61.5	57.8	52.5	49.0	46.6	
		MCDB	45.3	44.9	45.3	51.5	57.9	60.7	65.3	65.7	60.8	55.0	50.7	47.9	
5%		WB	42.9	42.1	42.9	45.6	50.3	54.0	59.0	59.6	56.5	51.3	47.4	44.9	
		MCDB	44.2	43.7	44.5	48.4	54.4	57.5	62.9	63.1	59.2	53.7	49.4	46.7	
10%		WB	41.5	40.7	41.4	44.2	48.5	52.5	57.4	58.3	55.4	50.1	45.6	43.1	
		MCDB	43.2	42.4	43.4	46.3	51.7	55.4	60.5	61.4	58.0	52.5	47.9	45.2	
Mean Daily Temperature Range	5% DB	MDBR	5.1	5.3	5.6	6.4	6.9	6.1	6.4	6.3	5.8	4.9	4.9	5.2	
		MCDBR	5.9	5.8	6.7	10.4	12.4	10.9	12.3	11.2	8.0	6.1	5.3	6.2	
		MCWBR	5.9	5.9	5.4	6.1	6.9	6.4	6.7	5.9	4.7	4.9	5.2	6.4	
	5% WB	MCDBR	5.8	5.7	5.9	9.0	11.1	9.6	11.1	9.8	7.1	5.5	5.3	5.8	
		MCWBR	6.0	6.1	5.5	5.9	6.8	6.3	6.8	5.7	4.8	5.0	5.6	6.3	
Clear-Sky Solar Irradiance	taub	0.391	0.319	0.341	0.361	0.355	0.356	0.364	0.355	0.342	0.316	0.361	0.360		
	taud	2.600	2.248	2.330	2.373	2.454	2.483	2.488	2.513	2.513	2.393	2.544	2.442		
	Ebn at Noon	99	188	237	259	272	274	268	260	239	191	98	43		
	Edh at Noon	9	19	25	30	30	30	30	26	22	16	8	5		
All-Sky Solar Radiation	RadAvg	39	185	567	1098	1496	1606	1484	1140	672	288	67	17		
	RadStd	6	31	80	92	74	156	134	112	82	43	12	2		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-1.53	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page